MIL-S-45578B(AL)
29 April 1985
SUPERSEDING
MIL-S-45578A(AL)
15 March 1977

MILITARY SPECIFICATION

SAW, POWER HACK, PORTABLE, PNEUMATIC

(2-inch to 8-inch pipe size capacity)

This specification is approved for use by the Armament Research and Development Center (Tool and Equipment Engineering Directorate, Standardization and Specification Engineering Division), Department of the Army, and is available for use by all departments and agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers one type of air-driven portable power hack saw.

2 APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-ETS (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3450

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SPECIFICATIONS

FEDERAL

WW-C-633 - Couplings, Hose: (Half), Pneumatic Universal Type
WW-P-404 - Pipe, Steel, (Seamless and Welded, Black and Zinc-Coated (Galvanized)
PPP-C-96 - Cans, Metal, 28 Gage and Lighter
GGG-B-451 - Blade-Hand Hacksaw and Power Hacksaw

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of
MIL-O-82042 - Filters and Oilers, Airline

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-889 - Dissimilar Metals

* 2.1.2 Other Government documents, drawings and publications. The following other Government documents form a part of this specification to the extent specified herein.

US DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29CFR 1910)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D3951-82 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3 REQUIREMENTS

3.1 First Article. When specified (see 6.2.1), the contractor shall furnish one or more saws for first article inspection. The first article may be either a preproduction model(s) or an initial production item(s) which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The saw shall be new and of the latest design meeting the requirements of this specification. All parts subject to wear or replacement shall be accessible. The saw shall be a pneumatic driven unit in a hand portable container.

3.2.1 Safety and health requirements. Covers or other safety devices shall be provided for all parts of the saw presenting safety hazards which can be guarded without substantial interference to operation of the saw. Except as otherwise specified by the procuring activity, the saw shall comply with the standards promulgated under OSHA 2206 applicable to the saw itself. All guards shall be removable to provide easy access to guarded parts. Exceptions and additional requirements for safety and health shall be as specified (see 6.2.1 and 6.3).

3.3 Construction. The saw shall be complete so that when installed it can be used for the operations specified herein.

* 3.4 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the power tool industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

* 3.4.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It is not permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

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* 3.4.2 Interchangeability. All replaceable parts shall be designed and manufactured to such standards as will permit replacement and adjustment without modification of parts or saw.

3.4.3 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No repair process shall be used for reclaiming any defective part.

3.4.4 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all requirements of the welded, the brazed, or soldered part. These operations shall not be employed as repair measures for defective parts.

3.4.5 Fastening devices. All screws, pins, bolts, and similar parts shall be installed in such a manner as to prevent loss of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

* 3.4.6 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

* 3.4.7 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with standard means for lubrication. Oil holes, grease fittings and filler caps shall be accessible without requiring disassembly of functional parts. Lubrication reservoirs, if applicable, shall have means for checking levels.

3.5 Performance and product characteristics.

3.5.1 Filter and oiler. An air line filter and oiler shall be attached to the saw ahead of the air motor and shall be in accordance with type I of MIL-F-82042.

3.5.2 Air control. A lever-type air control shall be provided on the air motor. The location of the air motor control shall permit the operation of the air control and the feed control at the same time by one person. Mechanical means shall be provided for holding the air control in the full ON position. The maximum speed rate shall not exceed 300 strokes per minute.

3.5.3 Blade holders. The frame shall be provided with means for holding the saw blade. A device shall be provided for adjusting the blade to proper tension and maintaining the blade in vertical alignment during its cutting operation. Special tools shall not be required for blade changing or adjustment.

3.5.4 Feed control. A positive manual pressure feed control shall be provided. The feed control shall be capable of regulating the cutting pressure through all phases of cutting.

3.5.5 Stroke. The length of the cutting stroke shall be not less than 2 inches and the blade shall lift from the cut on the return stroke.

3.5.6 Vise. The saw shall be equipped with a chain-type vise with a machined V-base or similar device for positive grip and alignment on the workpiece.

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3.5.7 Air motor. The air motor shall be rotary or piston type and shall have an air consumption of not more than 60 cubic feet per minute (cfm) free air at 100 pounds per square inch (psi) ±5 psi, inlet air pressure.

* 3.5.8 Saw blade. The saw shall be fitted with a saw blade meeting or exceeding GGG-B-451 for Type II, Grade A saw blades. The blade shall be not less than a nominal 1 inch in width.

3.5.9 Accessories. The saw shall be provided with 20 extra blades in accordance with 3.5.8, one standard male-female quick disconnect coupling conforming to WW-C-633, and one chain tensioning wrench. When specified (see 6.2.1), one pint of cutting oil, one pint of air motor oil, one suitable dispensing oil can, one 6-foot hose whip, and the tools that are required for making adjustments to the saw shall be provided.

3.5.10 Storage and carrying case. A rigidly constructed, portable, sheet metal storage and carrying case shall be furnished to completely house each saw and all accessories in a secured manner. The case shall have two handles for carrying purposes. A water and oil resistant canvas pouch or envelope or a plastic envelope of at least four mil thickness capable of holding two operating and maintenance manuals shall be securely fastened to the inside of the lid of the carrying case. The carrying case lid shall be equipped with closure devices that can be operated with a heavy wool-gloved hand. The carrying case shall be constructed to prevent entrance of rain when the case is in normal storage position.

* 3.6 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting all surfaces to be painted shall be clean and be free of all foreign matter detrimental to painting and at least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel) is applied.

* 3.7 Performance. The saw shall have the capacity to cut the following nominal pipe sizes: 2-inch, 4-inch, 6-inch, and 8-inch cast iron or steel pipe. The saw shall cut through a nominal size 6-inch, 3/8-inch thick wall steel pipe conforming to WW-P-404, in not more than 7 minutes, while meeting the accuracy requirements of 3.7.2.

3.7.1 Temperature and humidity conditions.

3.7.1.1 Low operating temperature. The saw shall meet the performance requirements of 3.7 while stabilized at an ambient temperature of -20 degrees Fahrenheit (F) to -25 degrees F.

3.7.1.2 High operating temperature. The saw shall meet the performance requirements of 3.7 while stabilized at an ambient temperature of +110 degrees F to +115 degrees F.

* 3.7.1.3 Low storage temperature. The saw shall be able to withstand storage without damage in a temperature of -60 degrees F to -65 degrees F for a period of 72 hours. Upon removal and stabilization at an air temperature of +75 degrees F to +85 degrees F, the saw shall conform to the performance requirements of 3.7.

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3.7.1.4 High storage temperature. The saw shall be able to withstand storage without damage in a 24-hour cycle of temperatures, starting with 10 hours at +85 degrees F to +95 degrees F, followed by 5 hours of steady increase to +150 degrees F to +155 degrees F, 4 hours at +150 degrees to +155 degrees F, and 5 hours of steady decrease to +70 degrees F to +90 degrees F. The saw shall conform to the performance requirements of 3.7.

3.7.2 Sawing accuracy. The accuracy of the saw shall be determined by the squareness of the cut to the longitudinal axis of a nominal size 6-inch steel pipe. This accuracy shall be within 0.03 inch out-of-square.

3.7.3 Operating position. The saw shall be a self-supporting unit, capable of operating within a trench 28 inches wide when clamped to a rigidly supported pipe by means of its vise, with the trench excavated 12 inches below the workpiece.

3.8 Weight. The weight of the saw shall not exceed 130 pounds and the weight of the carrying case and all accessory items shall not exceed 70 pounds.

3.9 Identification marking. The saw shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

3.10 Technical data. Data shall be furnished as specified (see 6.2.2).

* 3.11 Workmanship. Workmanship of the saw, accessories and auxiliary items shall be of a quality equal to that of the best commercial practices of the saw manufacturing industry.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).

b. Quality conformance inspection (see 4.2.2).

* 4.2.1 First article inspection. First article inspection shall be applied to the preproduction model or initial production items (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.3 and all tests under 4.4. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

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4.2.2 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production units offered for acceptance under the contract. Quality conformance inspection shall consist of a. through d. as follows and failure of the unit to pass either examination or any of the tests shall be cause for rejection.

a. Examination (see 4.3).

b. Cutting test (see 4.4.6).

c. Accuracy test (see 4.4.7).

d. Packaging inspection (see 4.5).

4.2.2.1 Sampling. When specified (see 6.2.1), sampling for lot acceptance shall be in accordance with MIL-STD-105 at the inspection level and acceptable quality level specified in the contract. Test samples for 4.4.6 and 4.4.7 shall be selected at random from the samples that passed the examination in 4.3.

* 4.3 Product examination. Visually and manually examine the power hack saw to determine conformance with the requirements of 3.2 through 3.6, 3.9 and 3.11. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, and proper cleaning. Manual examinations shall include the operation of moveable parts by hand to assure proper functioning. The saw, case, and accessory items shall be weighed in accordance with requirements of 3.8. Technical data shall be in accordance with requirements of 3.10. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the hack saw and its components have passed the examinations prescribed for by the contractor's inspection system shall be cause for rejection.

4.4 Tests.

4.4.1 Test conditions. Prior to performing the tests, the saw shall be serviced with lubricants in accordance with the technical data furnished with the saw.

4.4.2 Operating temperature tests. The complete saw shall be subjected to the low temperature conditions of 3.7.1.1 until stabilized. While at that temperature, the saw shall be performance tested in accordance with 4.4.6 and 4.4.7. The test shall be repeated for the high temperature conditions of 3.7.1.2.

4.4.3 Storage temperature tests. The complete saw shall be subjected to the low temperature conditions of 3.7.1.3, then allowed to stabilize in an air temperature of +70 degrees F to +90 degrees F. The saw shall then be inspected for evidence of damage, and performance tested in accordance with 4.4.6 and 4.4.7. The test shall be repeated for the high temperature conditions of 3.7.1.4.

* 4.4.4 Air control test. A suitable length of nominal size 6-inch, 3/8-inch thick wall steel pipe shall be secured in the saw. With the air control in the full ON position and the manual feed control at minimum position pressure, the number of blade strokes shall be counted by a suitable method (e.g., mechanically actuated counter) over a time interval of one minute. Blade speed in excess of 300 strokes per minute shall constitute failure of this test (see 3.5.2).

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* 4.4.5 Air consumption test. Connect a flow meter to the air discharge of the saw. Open the air-control valve and, while maintaining an air pressure of 100 psi +5 psi at the air inlet connection, determine the air consumption. Air consumption in excess of 60 cfm shall constitute failure of this test (see 3.5.7).

4.4.6 Cutting test. A 6-inch, 3/8-inch wall, nominal size steel pipe suitable for three consecutive cuts shall be secured in a work area to simulate a trench of not more than 28 inches wide (see 3.7.3). The operator shall comply with the technical data for setting up and operating the saw for a square cut. Not less than three cuts shall be made, utilizing the blade provided, in determining the cutting time. Actual cutting time only shall be used in determining the cutting rate and the average cutting rate shall not exceed 7 minutes (see 3.7).

4.4.7 Accuracy test. The last cut in 4.4.6 shall produce the piece to be checked for squareness. The surface of the cut shall not be out-of-square with the longitudinal axis in excess of 0.03 inch (see 3.7.2).

4.4.8 Sawing capacity test. One square cut shall be made in a suitable length of 2-inch and in a suitable length of 8-inch cast iron or steel pipe to verify the capacity of the saw to cut the range of nominal pipe sizes as specified in 3.7. Time and accuracy shall not be measured during this test.

4.5 Packaging inspection. Packaging shall be inspected to determine compliance with section 5.

5 PACKAGING

5.1 Preservation and packing. Preservation and packing shall be level A, B, or Industrial of MIL-M-18058, as specified (see 6.2.1). In addition, cutting oil and motor oil shall be stowed in metal containers conforming to PPP-C-96, type I, class 4.

5.2 Marking.

5.2.1 Military. When level A or B preservation and packing is specified, markings shall conform to MIL-STD-129.

* 5.2.2 Industrial. When industrial preservation and packing is specified, markings shall conform to ASTM D3951-82.

6 NOTES

6.1 Intended use. The power hack saw covered by this specification is a portable, air-motor-driven, reciprocating frame-type sawing machine intended to be clamped on the material to be cut, for use in trenches and general on-the-job cutting.

6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. First article, when required (see 3.1).

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c. Specify exceptions and additional safety and health requirements (see 3.2.1).

d. Accessories, when required (see 3.5.9).

e. Painting, if different (see 3.6).

f. Identification marking, if different (see 3.9).

g. First article inspection, if different (see 4.2.1).

h. Quality conformance inspection, if different (see 4.2.2).

i. Sampling for lot acceptance, when required (see 4.2.2.1).

j. Specify level A, B, or Industrial preservation and packing (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

* 6.3 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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